

Code Explanation Line by Line (Easy Language)

```
import numpy as np
```

- Imports NumPy library.
 - Used for handling arrays and numerical operations.
-

```
from sklearn.linear_model import Perceptron
```

- Imports Perceptron model from sklearn library.
-

```
import sys
```

- Imports system library.
 - (Not used in this code but imported.)
-

```
data = [  
    ('0', 48, 0),  
    ('1', 49, 1),
```

- Stores training data.
- Each tuple contains:
 1. Number character
 2. ASCII value
 3. Output class

Example:

- 0 → Even → 0
 - 1 → Odd → 1
-

```
def extract_feature(char_val):
```

- Creates a function named extract_feature.
-

```
    number = int(char_val)
```

- Converts character into integer.

Example:

- '4' → 4
-

```
    return [number % 2]
```

- Checks remainder after dividing by 2.

- If remainder:
 - 0 → Even
 - 1 → Odd

Example:

- $4 \% 2 = 0$
- $7 \% 2 = 1$

```
X_train = np.array([extract_feature(item[0]) for item in data])
```

- Creates input training data.

Generated values:

```
[[0],[1],[0],[1]...]
```

```
y_train = np.array([item[2] for item in data])
```

- Creates output labels.

Output:

```
[0,1,0,1...]
```

```
perceptron = Perceptron(max_iter=100, random_state=0)
```

- Creates perceptron model.
- max_iter=100 means training runs 100 times.
- random_state=0 gives same result every run.

```
print("Training the Perceptron...")
```

- Displays training message.

```
perceptron.fit(X_train, y_train)
```

- Trains the perceptron model using training data.

```
print("Training complete.\n")
```

- Displays completion message.

```
def recognize_even_odd(char_input):
```

- Function to recognize even or odd number.

```
if not char_input.isdigit():
```

- Checks whether input is digit or not.

```
return f"Input '{char_input}' is not a valid digit."
```

- Shows error if input is invalid.

```
feature = extract_feature(char_input)
```

- Extracts feature from input number.

```
prediction = perceptron.predict([feature])
```

- Predicts output using trained model.

```
status = "Odd" if prediction[0] == 1 else "Even"
```

- Converts prediction into text.

```
return f"The number {char_input} is recognized as {status}."
```

- Returns final result.

```
print(recognize_even_odd('4'))
```

- Tests number 4.

Output:

The number 4 is recognized as Even.

Expected Output

Training the Perceptron...

Training complete.

The number 4 is recognized as Even.

The number 7 is recognized as Odd.

The number 0 is recognized as Even.

The number 9 is recognized as Odd.

Viva Questions and Answers

1. What is Perceptron?

Perceptron is a simple neural network used for binary classification.

2. What is ANN?

ANN stands for Artificial Neural Network.

3. What is the use of Perceptron?

It is used to classify data into two classes.

4. What is binary classification?

Classification into two categories.

Example:

- Even/Odd
 - Yes/No
-

5. Which library is used for Perceptron?

`sklearn.linear_model`

6. Why NumPy is used?

For array handling and mathematical operations.

7. What is feature extraction?

Converting raw data into useful input format.

8. What does % 2 mean?

It checks remainder after division by 2.

9. How to identify even numbers?

If `number % 2 == 0`

10. How to identify odd numbers?

If `number % 2 == 1`

11. What is training in machine learning?

Teaching the model using data.

12. What is prediction?

Output generated by trained model.

13. What is fit() function?

Used to train the model.

14. What is predict() function?

Used to predict output.

15. What is ASCII value?

Numeric representation of characters.

Example:

- ASCII of 0 is 48.
-

Conclusion

Thus, we successfully implemented a Perceptron Neural Network to recognize even and odd numbers using Python and machine learning techniques.